

Cesium Fused Cohort Release Test Plan (S1 to S6)

Purpose: decide whether the 100 CVTE-fused Cesium tablets may be released, first to the ICON-controlled ring and then to customers. This plan executes Section 3.4 of the Cesium Hardware Root-of-Trust Authorization Plan r1.24. It does not authorize fusing of production units.

Product	Cesium tablet iFitG520, MediaTek MT8371 (tool reports MT8189), board C.G520.702, eMMC, battery-less
Units under test	Cohort of 100 fused by CVTE on 30 July 2026, currently on the VKC1_20260814 user build
Firmware note (r1.1)	The ring and test units must arrive in Logan on 0814, not reflashed. The 0814-to-0909 OTA is the first counted fused OTA transition (X1) and the package CVTE attached exists for it. S5 moves five ring units to 0909 with power interruptions; the rest of the ring takes the same OTA at ICON. The single S3/S4 unit reaches 0909 by depot flash instead, which is the other recovery path being proven. The 70 to 75 customer cohort units are different: production tablets flow ODM → console vendor → finished-goods vendor → warehouse or store and are not powered until a customer sets them up, so they must receive the 0814→0909 OTA at CVTE, logged per serial, before they leave CVTE, after S5 has passed on the ring. CVTE's lab reaches the iFIT OTA server (confirmed 16 Sep), so this is an OTA in the lab, not a flash. Customers then receive 0909, and the cohort's first field OTA coincides with the unfused control group's
Hold point	The customer cohort units are held at CVTE's outbound dock until the release decision below is signed. Once they enter the console vendor's flow they cannot be segregated by serial. The console vendor must record tablet serial against console serial at integration
Artifacts required	input.xml, GFH_CONFIG.ini, root_pubk.der (CVTE, 14 Sep); VKC1_20260909.zip containing signed DA_BR.bin and signed 0909 image; VKC1_20260814_20260909.zip OTA package; SP_Flash_Tool_Selector v1.2444 (use V6) with MediaTek USB VCOM drivers
Equipment	Windows PC with the VCOM drivers installed and a USB bus monitor (USBTreeView or equivalent); mini-USB cable; switchable or programmable 12 V supply; UART console lead for the DEBUG header; hand tools to open the enclosure
Unit location and executors	The 100 fused units are at CVTE. ICON holds the bricked unit and at most two other fused units. S1 and S2: ICON, desk work. S3 and S4: CVTE on a healthy fused unit, following the attempt A/B/C procedure below, with video of the USB bus monitor and tool console; ICON repeats if it has a healthy fused unit. S5: CVTE first, now, on five cohort units with UART logs and video (v1.3); ICON repeats on the ring units after they arrive. S6: CVTE, from the per-serial line logs plus three fresh read-backs. S7 and the US depot rehearsal: ICON, on the test units. CVTE ships the 15 to 20 ring units and the 7 test units now, on 0814; the shipment no longer gates the customer release
Owner / Test lead	Allen Middleton / Shane Andrus
Revision	1.9, 30 September 2026 (1.8 on 24 September; 1.7 and 1.6 earlier on 18 September; 1.5, 1.4, 1.3 and 1.2 on 17 September, 1.1 on 16 September, 1.0 on 15 September). v1.9: S3c added, ICON recovery of the first article by eMMC test point or in-system eMMC programming, not dependent on MediaTek. v1.8: S3b rewritten after CVTE reproduced the 15 Sep failure with Shane's preloader (main plan Source N); expected result is fail, run once for the record with UART; S1 record filled from CVTE's input.xml. v1.7: consequences of main plan r1.20 (MediaTek eFuse Writer Developer Guide V1.1 and Flashtool guide, Sources L and M): S1 records key_type, the hash slot, lock bits and Enable_SW_JTAG_CON against efuse_bingen.log; S3b and S4 record the SP Flash Tool scene and its NVRAM effect and enable "DL All with CheckSum" where checksum.ini exists; three dispositions of the external review revised. v1.6: dispositions of the external "v1.6" review against MediaTek Security 2.1 notes (table below); pbp.py made the reference for the S1 hash; Enable_SW_JTAG_CON recorded; Format all versus Download only caution in S3b; relay-harness automation and the PWR-01 to PWR-04 targeted subset added to S7; preloader UART evidence added. v1.5: Go/No-Go date 2 October; S7 is the confirmation condition on a Go and starts the day the test units arrive; Basecamp item 12 (line witness) added to the cross-reference. v1.4: release target 26 Sep; cross-reference to the 17 Sep Basecamp action items; 10 Oct fusing Go/No-Go noted. v1.3: S5 executor changed to CVTE first with ICON repeat; Malata early-start note. v1.2: S3 rewritten to CVTE's demonstrated 17 Sep procedure; S3b added
Target	Cohort release signed by 26 September if S1 to S6 pass. This plan does not decide production fusing; that is the 2 October Go/No-Go in the main plan (Option C, the M4 gate). S7 below is the confirmation condition on a Go and must start the day the test units arrive

Ordering rule. Run S1 and S2 first; they are desk work and either can stop everything. Run S3 before S4 on the same healthy unit. S5 and S6 can run in parallel with S3 and S4 on different units. S3b runs on the bricked unit only, at ICON, and can run today: it needs no cohort hardware.

Cross-reference to the 17 Sep Basecamp post (v1.4). The Basecamp message to CVTE, iFIT and Malata numbers actions per person. This table maps them to the cases here so results can be filed against the right step.

Basecamp item	Case here	Gates
Simon 1: ship ring and test units	Logistics; enables ICON's S5 repeat, S7, US depot rehearsal	Shipment
Simon 2: OTA five units with two power cuts	S5 (CVTE first)	Cohort release
Simon 3: depot flash a healthy fused unit	S4	Cohort release
Simon 4: read-efuse on three units, logs for 100	S6	Cohort release
Simon 5: recovery log, serial, versions	S3 evidence (ask 29)	Confidence
Simon 6: Format all vs Download only	Depot procedure (ask 30)	Confidence, H1
Simon 7: Force Flash is the BROM pin	Procedure (ask 32)	Confidence
Simon 8: Shane's method, then recover	Conflict 12 test, E5 second half (ask 31)	Confidence
Simon 9: MTK_SEC_BOOT, LK lock state	Feeds A0-b (ask 13)	Unfused fleet
Simon 10: AB_OTA_PARTITIONS, vendor-freeze	Feeds S2 (ask 27)	E1
Simon 11: OTA the customer units, hold at dock	Release step after S5	Cohort release
Simon 12: video-witnessed fuse-station run, week of 22 Sep	Main plan Tier 1, L1 to L8 and I1 to I6	Fusing Go/No-Go (Medium)
Shane 1: recover the bricked first article	S3b	E5, X3, confidence
Shane 2: fuse files and hash	S1	Cohort release
Shane 3: payload partition list	S2	Cohort release
Shane 4: unfused 0909 checks	A0-b (paper section)	Unfused fleet
Shane 5: ring placement, S5 repeat, soak, US depot	S5 repeat, S7 , H1	S7 confirms the 2 Oct Go; X2
Malata 1 to 4: unfused profile, experience, sacrificial units, channel	Main plan Tier 2, ME1 and ME4	Malata line
Malata 5 to 7: fuse, read back, brick and recover, boot	Main plan Tier 2, ME2 and ME3	Malata line

Status 24 Sep. CVTE's Force Flash procedure failed on a unit bricked with Shane's corrupted-signature preloader (main plan Source N, conflict 15). S3 remains passed for the wrong-key case only. S3b is expected to fail and is run once for the record. Cohort release is unaffected by the decision rule below: exposure remains boot0 storage failure, now understood as scrap rather than repair. CVTE has not yet reported S4, S5 or S6.

Status 17 Sep. CVTE has demonstrated S3 on a deliberately bricked fused unit (preloader signed to a key that does not match the burned hash) and recovered it. The procedure below is CVTE's, with the three conditions that differ from ICON's failed 15 Sep attempt marked **(condition)**. Shane's own brick method, requested by CVTE, is in S3b.

Dispositions of the external "v1.6" review (18 Sep). A separate analysis compared v1.5 with MediaTek Security 2.1 application notes and proposed a rewritten S1 to S7. It was written without the program record, so several of its "critical corrections" would undo decisions this plan made deliberately. Each is dispositioned here so the reasoning is on file.

Review recommendation	Disposition	Reason
Prohibit "Format all + download" on fused units; use "Download only" or "Firmware Upgrade"	Partly adopted; strengthened v1.7	"Format all" is the only BROM recovery demonstrated on this fuse map (Source J). It is not prohibited. The NVRAM and calibration concern is real, so S3b now records what "Format all" erased and ask 30 (does "Download only" also enter from BROM) decides the depot default. On the bricked first article nothing needs preserving. v1.7: the Flashtool guide (Source M) confirms "Format All" erases the NVRAM region and "Firmware Upgrade" restores it only from an ATE or SN Writer backup. S3b and S4 now record the scene used and check calibration afterwards; the depot preference order is in main plan H1.
Do not hash the raw .der; use pbp.py "PSS-padded outputs"	Adopted in part, wording softened v1.7	pbp.py is now the reference for SBC_PUBK_HASH in S1, with the manual hash as the cross-check. PSS is a signature padding scheme, not a hash input; the fuse holds a hash of the public key material in MediaTek's format, which is what pbp.py produces. v1.7: the eFuse Writer guide (Source L) shows the hash derivation depends on key_type (legacy or pss) in pl_key.ini, so the reviewer's instinct was right that a raw digest can differ from the fused value. pbp.py output or efuse_bingen.log is the reference; the raw digest is a cross-check that is expected to match only for legacy.
Enforce Enable_SW_JTAG_CON = 1 alongside SBC and DAA	Recorded, escalated v1.7	The cohort is already fused and its map is in the 27 July build log. A new flag means a new fuse image and belongs to the production-image review under G7 and G11, not to the cohort release. S1 records whether the key is present and its value. v1.7: Source L FAQ 5.2 says "make sure SW_JTAG_CON is blown" and the flag defaults true, but the 27 July build log does not list it. S1 now records its value from input.xml, efuse_bingen.log and the read-back; the production image decision is main plan G7 and ask 34. The cohort is not re-fused either way.
S1 pass criterion "DAA = 1"	Rejected	DAA stays unblown by decision (G11, STOP-SHIP ; conflict 8). Blowing DAA makes every download require an AuthFile the depot chain does not have and turns Source E's worst-case recovery into Cesium's. Read-back already shows DAA off
S2 "eFuse programming on 100 units, SP Flash Tool or preloader self-blow, V_BAT above 3.7 V"	Rejected	The 100 are already fused. Self-blow is ruled out (Source C, Section 0 item 1). Cesium has no battery. This item shows the review did not have the program state
S4 "flash signed production firmware onto all 100 via Download only"	Rejected	The cohort moves 0814 to 0909 by OTA on purpose: it is the first counted fused OTA transition (X1) and the S5 power-cut cases depend on it. Flashing would discard the evidence
S5 "Virtual A/B COW snapshot, snapshotctl rollback"	Not assumed	Whether Cesium uses virtual A/B is unconfirmed (conflict 2, M2 and M3). S5 is written to be correct for plain A/B or virtual A/B
S6 "UART audit of all 100 units during S5, efuse blow: success"	Adopted as supplementary evidence	The preloader log line that matters is hw_sbc: 1 and the signature-check pass (Source I). It is captured on every unit that has a UART lead attached (S5 units 4 and 5, S3b, S7) and filed with the read-back. It does not replace read-efuse sampling: UART needs the enclosure open on each unit and the 100 are at CVTE
S7 relay harness and PWR-01 to PWR-04 targeted cuts	Adopted	Programmable relay control was already the S7 setup. The four targeted phases from Source A are added as a structured subset within the 200-cycle random soak, so both the specific windows and the random distribution are covered
"Total execution time 3.5 hours"	Rejected	One 436 MB OTA is tens of minutes per unit; 200 soak cycles are about a week unattended; S7 is the confirmation condition on the 2 October decision precisely because it takes that long. A release plan sized to 3.5 hours has removed the test that measures brick risk
Release on "100 percent pass across S1 to S7"	Not adopted	The release decision table below is the rule: S3 and S3b outcomes do not gate the cohort release, S7 gates the fusing Go, and each failure mode has its own consequence

Global record per unit. Serial, board revision, firmware fingerprint before and after, tester, date, tool and driver versions, console log file name, photos where the step says so.

S1. Fuse image contents and independent hash check

Gates: P1, G4, G7. **Effort:** 1 hour. **Units:** none (desk work) plus the read-back log from Shane's 15 Sep unit.

Checklist

- ☐ Open `input.xml` from CVTE. Record the value of every enable switch present. Expected keys per MediaTek Secure Boot Developer Guide V1.1 section 7: `Enable_SBC`, `Enable_DAA`, `Disable_Rom_Cmd`. Record any additional keys verbatim (for example `Disable_JTAG`, `Enable_SW_JTAG_CON`, `SBC_PUBK_HASH` and its value).
- ☐ Open `GFH_CONFIG.ini`. Record `brom_magic_cmd_mode_permanent_dis`, `jtag_en`, `debug_en`, `sec_level`, and any key containing `dis`, `lock` or `perm`.
- ☐ **Reference method (v1.6):** run MediaTek's `pbp.py` from the eFuse Writer package against `root_pubk.der` and record the `SBC_PUBK_HASH` it emits. This is the value the fuse image was built from. **Cross-check:** compute `openssl dgst -sha256 root_pubk.der` and, if that does not match, the SHA-256 of the modulus alone; record which form equals the `pbp.py` output so the derivation is documented for Malata and the depot. If `pbp.py` is not available at ICON, ask CVTE to run it and send the output with the command line.
- ☐ **v1.7:** obtain `efuse_bingen.log` from the 27 July build (ask 34) and the `key_type` line from `pl_key.ini`. Record which `SBC_PUBK_HASH` slot the image writes and the state of the other slots and of every `Disable_SBC_PUBK_HASHn` bit. Record every lock bit (`sec_attr_lock`, `sbc_pubk_hash_lock` and the rest) from the read-back; expected unlocked per Shane's 15 Sep read-back. Record `Enable_SW_JTAG_CON` from the log, not only from `input.xml`.
- ☐ Compare the computed hash to `sbc_pub_key_hash` in Shane's 15 Sep `read-efuse` output and to the hash embedded in `input.xml`.
- ☐ File both configuration files, the `read-efuse` log and the computed hash in the test record.

Record

Field	Value
<code>Enable_SBC</code>	true (CVTE input.xml, 18 Sep)
<code>Enable_DAA</code>	false (CVTE input.xml, 18 Sep)
<code>Disable_Rom_Cmd</code>	
<code>brom_magic_cmd_mode_permanent_dis</code>	
<code>jtag_en</code> / <code>debug_en</code>	
<code>Enable_SW_JTAG_CON</code> (present? value)	false (CVTE input.xml, 18 Sep)
<code>key_type</code> in <code>pl_key.ini</code> (legacy / pss)	
Hash slot written; other slots and disable bits	
Lock bits set (expected none)	
<code>efuse_bingen.log</code> on file (date, hash it reports)	
Hash computed by ICON	
Hash from <code>read-efuse</code>	
Hash in <code>input.xml</code>	

Cross-check (received 16 Sep). CVTE's build log for `efuse_iFitG520.img`, dated 27 July 2026, reports `EFUSE_Enable_SBC = 1`, `EFUSE_Enable_DAA = 0`, `EFUSE_Enable_SLA = 0`, `EFUSE_Disable_BROM_CMD = 0`, `EFUSE_Disable_DBGPORT_LOCK = 0`, `EFUSE_USB_download_type = 0`, and states `brom_magic_cmd_mode_permanent_dis` does not exist in the project. The values read from `input.xml` must agree with this log; a disagreement is itself a finding.

Pass: `Enable_SBC` true; `Enable_DAA` false; `Disable_Rom_Cmd` (or `Disable_BROM_CMD`) false and `brom_magic_cmd_mode_permanent_dis` 0 or absent; values agree with CVTE's 27 July build log; ICON-computed hash equals the read-back hash. **v1.7**: hash in `efuse_bingen.log` equals the read-back; no lock bit set unless G12 has recorded a decision to set it.

Fail: any of: `Disable_Rom_Cmd` true or `brom_magic_cmd_mode_permanent_dis` 1 (stop: no cohort unit to customers, fuse image must be regenerated); `Enable_DAA` true (stop: recovery requires an AuthFile that does not exist in the depot chain); hash mismatch (stop: the key CVTE sent is not the key that was burned, escalate before anything else).

S2. OTA payload partition scope

Gates: E1. **Effort:** 1 hour. **Units:** none (desk work).

Checklist

- ☐ Extract `payload.bin` and `payload_properties.txt` from `VKC1_20260814_20260909.zip`.
- ☐ List the partitions in the payload manifest. Any AOSP payload dumper works (for example `python3 payload_dumper.py --list payload.bin`); the manifest is protobuf at the head of `payload.bin`. If no dumper is available, `strings payload.bin | head -200` shows the partition names in the manifest.
- ☐ Repeat for the full 0909 package from `VKC1_20260909.zip` if it contains an OTA payload (the factory image set is not an OTA and is out of scope here).
- ☐ Record every partition name. Mark each as A/B slot partition (boot, system, vendor, product, vbmeta and variants, dtbo, lk, tee, scp, spmfw, and so on) or as boot0 / preloader.
- ☐ Ask CVTE for `AB_OTA_PARTITIONS` from the 0909 board configuration (`BoardConfig.mk` / `device.mk`) and record it. If `preloader` is absent from that list the OTA builder cannot emit a preloader image at all, which is the stronger form of the E1 guarantee. Record whether CVTE describes the Android 15 BSP as a vendor-freezie release.

Record

Partition	Present in 0814 → 0909 payload	Class
preloader / preloader_a / preloader_b / boot0		non-A/B, BROM-verified
lk / lk_a / lk_b		A/B, preloader-verified
tee / tee_a / tee_b		A/B, preloader-verified
boot, vbmeta*, dtbo, system*, vendor*, product*, odm*, *_dlkm		A/B
other (list)		

Pass: no preloader or boot0 partition in any OTA payload. LK, TEE and Android partitions may be present: a bad LK is refused by the fused preloader, which then offers preloader USB download mode, so

that failure is depot-recoverable.

Fail: preloader or boot0 present in an OTA payload. Stop. This is a release-pipeline change before either the cohort or the unfused MP fleet ships, because a corrupt preloader on a fused unit is the one hard-brick mode and the same package would go to the unfused fleet.

S3. Forced download entry on a healthy fused unit

Gates: P2 first half, conflict 11. **Effort:** 1 hour. **Units:** one healthy cohort unit that boots normally. **Status:** passed at CVTE on 17 Sep on a bricked unit, which is the stronger case. ICON runs it on a healthy unit only for the record and to rehearse S3b; if no healthy fused unit is at ICON, skip to S3b.

Setup checklist

- ☐ Confirm the unit boots to Android and record `getprop ro.build.fingerprint`, `ro.boot.verifiedbootstate`, `ro.boot.veritymode`.
- ☐ Power off. Open the enclosure. Photograph the board and identify the **Force Flash** tact switch CVTE marked in the 14 Sep and 17 Sep photos (near the `DC_12V_IN` header). Confirm against CVTE's photo that it is Force Flash and not `RESET`.
- ☐ Install MediaTek USB VCOM drivers on the PC. Confirm in Device Manager that no MediaTek device is currently listed.
- ☐ Start SP Flash Tool V6. Load `flash.xml` from the 0909 package. Connection USB, signed `DA_BR.bin` selected. **(condition 1)** Set the scene to **Format all + download**, exactly as CVTE did. For S3 on a healthy unit this scene would erase the unit, so the tester must be ready to cancel at the handshake; if the tool version cannot be cancelled cleanly, run S3 only on a unit that S4 will reflash anyway.
- ☐ **(condition 2)** Click **Download** now, before power, so the tool is actively polling for a BROM device. Confirm the status bar shows it waiting.
- ☐ Start the USB bus monitor with a capture log. Note the time.
- ☐ Connect the mini-USB cable to the PC. 12 V still off.

Attempt A: BROM window without key (reference only)

- ☐ Apply 12 V with no button pressed. Watch the bus monitor for 30 seconds. Record any device that appears, with VID:PID and how long it stays. Expected candidates: `0E8D:0003` (BROM), `0E8D:2000` (preloader). Remove 12 V. If the tool starts a download, cancel it.

Attempt B: Force Flash held before power (CVTE's procedure)

- ☐ **(condition 3)** Press and hold Force Flash **first**, with 12 V still off. Then apply 12 V while continuing to hold. Keep holding until the tool reports a device (CVTE: "ensure it is low after power-on"). Do not release at a fixed time.
- ☐ Record: devices seen with VID:PID, the tool console from first contact, and whether the tool reports the BROM or the DA stage. Once the tool has connected and identified the device, **cancel** before any partition is written. Remove 12 V.

Attempt C: repeat of B (repeatability)

- ☐ Power-cycle and repeat Attempt B once. Two successes out of two are required, because the 15 Sep failure was a single attempt with no repeat.

Record

Attempt	Devices seen (VID:PID)	Time from 12 V to first enumeration	Tool stage reached (BROM / DA / download started)	Console excerpt
A: no key				
B: Force Flash held before power				
C: repeat of B				

Pass: in Attempts B and C, a MediaTek device enumerates and SP Flash Tool reports a connection (BROM handshake or DA loaded). Record which mode by VID:PID; `0E8D:0003` is the BROM entry that E5 needs. The unit must boot normally afterwards.

Fail: no MediaTek device enumerates in B or C with all three conditions met. Interpret with S1 (ROM command disable state) and with CVTE's 17 Sep result: since CVTE's unit entered BROM on the same fuse map, a failure at ICON is a procedure or fixture difference (driver, cable, hub, button contact, 12 V rise time) before it is a BROM behavior question. Photograph the setup and send CVTE the console log.

Note on Attempt A. If the preloader enumerates briefly even with no key, that is the normal preloader USB window used by the factory SOP and is sufficient for S4. It says nothing about BROM entry when the preloader is bad, which is the E5 question; only a BROM-mode device (`0E8D:0003`) in Attempt B or C answers that.

S3b. BROM recovery of the bricked first article J26080143-0A00076

Gates: E5, X3, P2 second half. **Effort:** 1 hour. **Units:** the bricked first article (preloader RSA signature corrupted in boot0 on 16 Sep, boot1 intact, fused 8 Sep, userdebug VKC1_20260907, unlocked). **Runs at ICON today; no cohort hardware needed.**

Why this should work now (v1.2, superseded). CVTE's 17 Sep brick and this one are the same BootROM event: BROM rejects the preloader in boot0 against the burned hash, whether the signature is corrupted (this unit) or valid under the wrong key (CVTE's). CVTE recovered theirs. The two differences from the 15 Sep attempt on this unit are the button already low when 12 V is applied and the tool already polling in "Format all + download". One difference between the units is untested: this one was fused first and corrupted afterwards; CVTE's was flashed with mismatched firmware and then fused.

Why it is now expected to fail (v1.8). On 18 Sep CVTE flashed this exact corrupted preloader onto one of its own fused units and could not recover it with the procedure that worked on 17 Sep. The wrong-key and corrupted-signature cases behave differently in BROM (main plan conflict 15), and MediaTek has been asked why. Run S3b once with UART attached so the record has ICON's own console capture; do not repeat beyond two attempts, and do not brick any further unit until MediaTek answers.

Checklist

- ☐ Open the enclosure and locate Force Flash per the S3 photo. Check the switch with a meter: it must pull the line low when pressed and the contact must hold (a worn tact switch that bounces open during the BROM sample is a plausible cause of the 15 Sep result).
- ☐ SP Flash Tool V6, `flash.xml` from `VKC1_20260909.zip`, signed `DA_BR.bin`, connection USB, scene **Format all + download**, exactly as CVTE demonstrated. Note this erases userdata and may erase NVRAM or calibration data; nothing on the first article needs preserving, but **record which partitions the scene formatted** (tool log) so the depot knows the cost of this path. **v1.6:** do not

substitute "Download only" on the bricked unit until CVTE answers ask 30; the proven procedure comes first. If ask 30 confirms "Download only" also enters from BROM, it becomes the depot default and "Format all" the fallback. Confirm every image in the scene is the signed 0909 set, since BROM will verify the DA and the new preloader against the burned hash.

- ☐ **v1.7:** record the scene used. If the unit enumerates, try **Download Only** first with the same package and only fall back to **Format all + download** if the DA refuses or the write fails (Source M: Format All erases NVRAM; Download Only touches only the partitions it writes). Either result answers ask 30 for this unit. If the package includes `checksum.ini`, enable "DL All with CheckSum" in the DA options. If the BROM does not enumerate on the first hold, repeat the hold several times before changing anything else (Source M FAQ on `Kcol0`).
- ☐ Click **Download** so the tool is listening. Start the USB bus monitor. Connect the cable. 12 V off.
- ☐ Press and hold Force Flash. Apply 12 V while holding. Keep holding until the tool reports a device or 60 seconds pass. Record the time from 12 V to first enumeration and the VID:PID.
- ☐ Let the download run to "Download Ok". Record the console log in full.
- ☐ Power-cycle. Confirm boot to Android and record `ro.build.fingerprint` (expected VKC1_20260909), `ro.boot.verifiedbootstate`, `ro.boot.veritymode`. **v1.6:** with the UART lead attached, capture the preloader boot log and file the `hw sbc: 1` and signature-check lines as corroboration of the fuse state.
- ☐ `read-efuse`. Compare to the 8 Sep read-back: `sbc_en` on, hash `6da1756f...83730c68`, DAA, SLA, JTAG-disable off. A flash must never alter fuse state.
- ☐ If no enumeration after two attempts: photograph the setup, capture the bus-monitor log, and try once with a different cable and a USB 2.0 port on a different host. Then stop and send CVTE the logs with ask 31; do not proceed to the eMMC test point without CVTE's board guidance (ask 24).

Record

Field	Value
Attempt count to first enumeration	
VID:PID at first enumeration	
Time from 12 V to enumeration	
Tool result	
Fingerprint after boot	
verifiedbootstate / veritymode	
read-efuse identical to 8 Sep	

Pass: the unit enumerates, "Download Ok", boots VKC1_20260909 with verified boot green, fuse read-back unchanged. E5 and X3 close; confidence becomes eligible for Medium per the main plan Section 3.5.

Fail: no enumeration with all conditions met and two cable/host variations tried. Confidence stays Low. The order-of-events difference (fused-then-corrupted versus mismatched-then-fused) becomes the question for CVTE and MediaTek (ask 31), and the eMMC test point (ask 24) returns as the fallback path.

Method given to CVTE (their request of 17 Sep). Binary delivered 18 Sep: `preloader_iFitG520.bin` (corrupted copy), 736,868 bytes, SHA-256

`b78fb3c34bc8d6589dbe52f50c3c287a693a1bd943cb431012f994e93f024980`, 32 bytes at offset `0xB3C28` XOR `0xFF` on the VKC1_20260907 signed preloader; flashed to boot0 only, Download Only scene.

Delivered the same day as the full G520 eFuse Burn Log PDF. Summary for the record: on a fused unit that boots, take the signed production `preloader.bin`, flip 32 bytes inside the RSA signature block (leave the code and the GFH header untouched), write it to boot0 (`preloader_a`) only with SP Flash Tool

"Download only", leave boot1 intact, power-cycle. Record whether the tool accepts the write; on 16 Sep it did, which is conflict 12. Then attempt the same BROM recovery.

S3c. Recovery of the first article without BootROM download (v1.9)

Gates: E5, X3 (corrupted-signature case). **Effort:** half a day once pad locations are known. **Units:** J26080143-0A00076. **Precondition:** eMMC pad locations from CVTE (main plan ask 41), or found with a scope.

Step 0, UART capture. Attach the UART lead, power on, capture 60 seconds. File whatever BROM prints, even nothing. Send to CVTE for MediaTek.

Route A, eMMC test point to BROM download. - [] Identify eMMC CLK or DAT0 on the board (pad, or the board-side end of a series resistor). Confirm with a scope on a healthy unit: CLK shows a burst within the first 100 ms of 12 V. - [] SP Flash Tool V6, signed `DA_BR.bin`, `flash.xml` from VKC1_20260909, scene Download Only, Download clicked and waiting. USB bus monitor running. - [] Short the chosen line to ground with a fine probe. Hold Force Flash. Apply 12 V. Watch for enumeration (0E8D:0003). Once the tool reports the BROM handshake or DA loaded, release the short. - [] Let the download write the preloader. Power-cycle. Record boot, fingerprint, `read-efuse` unchanged. - [] Two attempts maximum per line; if no enumeration, try the other line once, then stop.

Route B, in-system eMMC programming. - [] eMMC ISP programmer (UFI, Easy JTAG or Medusa class) wired to CLK, CMD, DAT0, GND, with VCC and VCCQ supplied per the programmer's guidance; tablet's own 12 V off. - [] Read boot partition 0 first and save it; confirm the corrupted preloader is present (compare to the r1.21 artifact's hash) so the procedure is proven against the real fault. - [] Write the signed production `preloader_iFitG520.bin` (VKC1_20260909 package) into boot partition 0. Read back and compare. - [] Disconnect, power on. Record boot, fingerprint, `read-efuse` unchanged, preloader log `hw sbc: 1`.

Pass: unit boots signed 0909 by either route; fuse read-back unchanged. E5 and X3 close for the corrupted-signature case. Route B success is additionally recorded as the candidate depot procedure for boot0 corruption (H1, R19).

Fail: no enumeration in Route A and Route B unavailable or unsuccessful. The unit is preserved for MediaTek. Confidence stays Low; the 2 Oct decision is No-Go by rule.

S4. Depot flash of a healthy fused unit with signed artifacts

Gates: H1 in cohort scope. **Effort:** 1 hour. **Units:** the same unit as S3, after it has been confirmed to boot.

Checklist

- ☐ Before flashing: `read-efuse` and save the log. Record `sbc_en`, `sbc_pub_key_hash`, `daa_en`, `sla_en`, `jtag_dis`.
- ☐ SP Flash Tool V6, `flash.xml` from VKC1_20260909.zip, signed `DA_BR.bin`, connection USB. Choose the **Firmware Upgrade** profile (the boot-chain profile the SOP documents). **v1.7:** if `checksum.ini` is in the package, enable "DL All with CheckSum" so the DA verifies each image after writing (Source M); note in the record whether the option was available.

- ☐ Enter download mode by whichever method passed in S3 (or the plain preloader window from Attempt A). Run the download. Record the console log through `Download 0k` or the failure message.
- ☐ Power cycle. Confirm the unit boots to Android. Record fingerprint (`VKC1_20260909`), `ro.boot.verifiedbootstate`, `ro.boot.veritymode`.
- ☐ `read-efuse` again. Compare to the pre-flash log.
- ☐ If a **Download Only** or full-image profile exists in the package, repeat the flash with it and record the partition list it wrote. This is the profile the depot needs for failures outside the boot chain (R19).
v1.7: after each scene, confirm Wi-Fi MAC, Bluetooth address and any touch or display calibration are intact; a loss points at the NVRAM region (Source M) and must be written into the depot procedure as a restoration step.

Record

Field	Before	After
Fingerprint		
<code>sbc_en</code> / hash		
<code>daa_en</code> / <code>sla_en</code> / <code>jtag_dis</code>		
<code>verifiedbootstate</code> / <code>veritymode</code>		
Profile used and result		
<code>checksum.ini</code> present / CheckSum enabled		
NVRAM and calibration intact after flash		

Pass: `Download 0k`; boots to 0909; fuse read-back byte-identical before and after; verified boot state green and verity enforcing.

Fail: tool refuses the DA (indicates DA authentication is on despite `daa_en` off, escalate); download completes but the unit does not boot (record console, do not retry until the log is reviewed); fuse read-back changed (stop everything; a flash must never alter fuse state).

S5. First cohort OTA transition with power interruption

Gates: X1 first transition, D2 and D4 in cohort scope, P6. **Effort**: 1 day. **Units**: five healthy cohort units on VKC1_20260814. Two of the five are the interruption units.

Executor (v1.3). CVTE runs S5 first, at CVTE, now. Evidence to send ICON: `update_engine` log per serial, UART console capture on units 4 and 5 through the cut and the reboot, fingerprint before and after for all five, and video of the two power cuts showing the progress indication at the moment of the cut. ICON accepts a CVTE pass as the S5 result for the customer-release decision. ICON repeats S5 on five ring units when they arrive in Logan, as the independent check; the repeat feeds X2 and the confidence level, not the release. Where a step below says "trigger the update", CVTE uses the iFIT OTA server from its lab, which it can reach; sideload is the fallback and must be recorded if used.

Malata (v1.3, for information). Malata is not asked to run S5 or S7. Malata's early-start items are in the main plan Tier 2 (ME1 to ME4): controlled receipt of the fuse image and DA, three to five sacrificial units fused and read back, an independent brick-and-recover using Shane's method and CVTE's Force Flash procedure, and the unfused factory-test profile.

Setup checklist

- ☐ Record the fingerprint and active slot (`bootctl get-current-slot` via adb if enabled on the user build, otherwise from the boot console) for all five.
- ☐ Stage `VKC1_20260814_20260909.zip` on the OTA server ring reserved for the cohort, or sideload it via recovery if the backend is not ready. Record which path was used; it must be the same path the MP fleet will use for X2 to be meaningful later.
- ☐ Attach the UART console to the two interruption units. Use the switchable 12 V supply for those two. **v1.6:** where a programmable relay is available, script the two cuts (install progress 40 to 60 percent; first boot after kernel start) instead of pulling the cable, and log the cut timestamp against the console. Capture the preloader `hw sbc: 1` line on each reboot as supplementary fuse evidence.

Units 1 to 3: clean update

- ☐ Trigger the update. Record download start and end, install progress messages, slot switch, reboot.
- ☐ After reboot: fingerprint is 0909, active slot changed, boot-success marker set (`bootctl is-slot-marked-successful` or the console equivalent), `verifiedbootstate` green, `veritymode` enforcing.
- ☐ Power cycle once more and confirm it stays on 0909.

Unit 4: power cut during payload write

- ☐ Trigger the update. When install progress is between 40 and 60 percent (console or update log), remove 12 V.
- ☐ Wait 10 seconds. Reapply 12 V. Record which slot boots and the fingerprint. Expected: the old slot, 0814, boots normally.
- ☐ Let the update resume or restart. Record whether it resumes from checkpoint or restarts from zero. Complete the update. Confirm 0909 boots with success marker set.

Unit 4b (optional, if LK or TEE appear in the S2 partition list): power cut during the LK/TEE write.

`update_engine` writes partitions in manifest order; watch the console for the `lk` or `tee` partition name and remove 12 V while it is being written. Expected: identical to unit 4, the old slot boots because the slot switch has not happened. This is the one write whose failure a fused preloader would refuse rather than crash on, and it is the narrowest window in the whole OTA.

Unit 5: power cut during first boot of the new slot

- ☐ Trigger the update and let it complete to the reboot. On the first boot of the new slot, remove 12 V after the kernel starts and before Android reaches the home screen (roughly when the boot animation appears).
- ☐ Reapply 12 V. Record which slot boots, fingerprint, and the boot-attempt counter from the console. Repeat the cut up to three times if the new slot keeps being selected, so the retry budget is exercised.
- ☐ Final state: either 0909 boots and marks successful, or the bootloader falls back to 0814 after the retry budget. Record which.

Record

Unit	Path (OTA/sideload)	Interruption	Slot before → after	Fingerprint after	Success marker	Boot attempts	Outcome
1		none					
2		none					
3		none					

Unit	Path (OTA/sideload)	Interruption	Slot before → after	Fingerprint after	Success marker	Boot attempts	Outcome
4		40–60 % write					
5		first boot					

Pass: all five end on a booting build. Units 1 to 3 on 0909 with success marker set. Unit 4 boots the old slot after the cut and then completes to 0909. Unit 5 ends on 0909 or falls back to 0814 within the retry budget. No unit requires the depot.

Fail: any unit that does not boot either slot after power is restored (record everything, attempt S4 recovery on it, and treat it as a cohort stop until root-caused); any unit that boots but reports verified boot state other than green; the update engine refusing to start on a battery-less unit (record the reason; this is the N6 health HAL concern and blocks the transition for the whole cohort).

S6. Fuse read-back sampling

Gates: P1 sampling. **Effort**: 1 hour. **Units**: three cohort units chosen at random from serials not used in S3 to S5.

Checklist

- ☐ Pick three serials by a method that is recorded (random number against the serial list).
- ☐ For each: preloader-mode USB connection, SP Flash Tool read-efuse, save the log against the serial.
- ☐ Compare each to the S1 reference: `sbc_en`, `sbc_pub_key_hash`, `daa_en`, `sla_en`, `jtag_dis`, and the ROM command-disable bit if the tool reports it.
- ☐ Confirm each boots normally afterwards.

Record

Serial	sbc_en	Hash matches S1	daa_en	sla_en	jtag_dis	ROM cmd disable	Boots

Pass: all three identical to the S1 reference and booting.

Fail: any unit with a different hash or a different fuse map. Stop sampling, read back ten more, and treat the cohort as two populations until CVTE's per-serial line logs explain the difference.

S7 (required, starts the day the seven test units arrive). Randomized power-interruption soak on fused units

Gates: D7 in cohort scope; answers the power-loss question directly. **v1.5: S7 is the confirmation condition on a 2 October Go for fusing lot 1**. It is the one test that measures brick risk after the M4 gate. Units arrive about 25 September; 200 cycles take about a week; result due about 5 October, before the line runs. Zero unrecoverable confirms the Go. Any unrecoverable unit converts it to No-Go and lot 1 ships unfused. **Effort**: about one week unattended. **Units**: the seven test units (CLOSED-1..7), or as many as remain healthy after S3/S4.

Setup. Programmable 12 V supply or relay harness under script control; UART console logging on each unit; a repeatable OTA source that can be re-armed (the 0814 → 0909 package applied to a unit that is then flashed back to 0814 by the S4 depot path, or two signed builds that can ping-pong between slots).

Structure (v1.6). The 200 cycles are split: 160 with the cut at a uniformly random point across download, install and first boot, and 40 targeted at the four windows from the A/B resiliency plan (Source A), ten each. The targeted set proves the specific windows; the random set measures the distribution.

ID	Window	Cut	Expected
PWR-01	Early boot, preloader and LK, first 500 ms after 12 V	During bootloader handshake	Restarts cleanly; active slot retained
PWR-02	Payload write to the inactive slot (or COW snapshot if virtual A/B)	During <code>update_engine</code> write	Update aborts or resumes; active slot boots; no data corruption
PWR-03	Slot-switch commit	During the boot-control flag update	Atomic metadata write; boots a known-good slot, never an ambiguous state
PWR-04	First boot of the new slot	Before <code>boot_completed</code>	Retry budget, then fallback to the old slot; on Cesium the AVB rollback index is 0 and inert, so this is a retry-budget test, not a rollback-index test

Checklist

- ☐ Script: start OTA, wait a uniformly random interval across the full download-plus-install-plus-first-boot window, cut 12 V for 5 seconds, restore, wait for boot, record outcome and the OTA phase at the cut from the console.
- ☐ Run 200 cycles spread across the units: 160 random, 40 targeted (PWR-01 to PWR-04, ten each). Record per cycle: phase at cut, slot before and after, boot attempts, final fingerprint, preloader `hw sbc: 1` seen, recoverable yes/no.
- ☐ Any cycle that does not end in a booting unit: stop that unit, preserve console log, attempt S4 recovery, root-cause before continuing.

Pass: zero unrecoverable units over 200 cycles. Failure distribution by phase reported.

Fail: any unrecoverable unit. Confidence drops to Not ready until root-caused.

Non-test preconditions (paper)

These are not tests but the release decision needs them on file.

- ☐ **P4 key decision.** Written: no rotation for the cohort or MP; residual risk of the Basecamp-distributed private keys accepted; attachments removed from Basecamp; holder named for `root_prvk.pem`, `img_prvk.pem`, `da_prvk.pem`, and the Android platform keys.
 - ☐ **P5 backend records.** Every cohort serial recorded in the OTA backend with fuse state = fused, fingerprint, slot, and a depot routing flag that prevents any unsigned flash attempt.
 - ☐ **A0-b on one unfused 0909 unit.** `ro.boot.verifiedbootstate=green`, `ro.boot.flash.locked=1`, fastboot flashing unlock refused, and a wrong-key vbmeta refused at boot. Pass means the unfused MP fleet enforces AVB from LK and ships as built. Fail means the preloader flag change is required before 2 October.
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Release decision

S1	S2	S3	S4	S5	S6	Paper	Decision
Pass	Pass	Pass	Pass	Pass	Pass	Done	Release ICON ring now; release customer units from CVTE's dock after they have taken the 0814-to-0909 OTA at CVTE with per-serial logs
Pass	Pass	Fail	Pass	Pass	Pass	Done	Release ICON ring; hold customers for MediaTek's answer to ask 15. Exposure meanwhile is boot0 storage failure only, since S2 shows OTA cannot write the preloader. Not the active branch from 17 Sep: S3 passed at CVTE
any	any	Pass (CVTE)	any	any	any	any	S3b result does not change the cohort release. S3b pass closes E5 and X3 and moves confidence toward Medium (fused production); S3b fail keeps confidence Low and reopens ask 15 with the order-of-events question
Fail: ROM cmd disable set	any	any	any	any	any	any	No cohort unit to customers. The 100 stay ICON-internal. Fuse image regenerated before any further fusing
Fail: hash mismatch or DAA on	any	any	any	any	any	any	Stop; escalate to CVTE before any other step
any	Fail	any	any	any	any	any	Stop; release pipeline must exclude preloader before cohort or MP ships
any	any	any	Fail	any	any	any	Stop; no depot path for fused units; root-cause before release
any	any	any	any	Fail	any	any	Stop; the A/B path failed on a fused unit; root-cause with console logs
any	any	any	any	any	Fail	any	Widen sampling; cohort treated as two populations

Sign-off

Role	Name	Date	Signature
Test lead	Shane Andrus		
Owner	Allen Middleton		
Approver (release of cohort to ICON ring)			
Approver (release of cohort to customers)			

This plan releases the fused cohort only. Authorization to fuse production units follows the full Cesium Hardware Root-of-Trust Authorization Plan, Section 4.